

MAGNOLIIDS

The magnoliids are a large group of primitive flowering plants that in botanical terms are between basal angiosperms and monocots.

Found in tropical and temperate regions, the magnoliids form a major plant group that emerged early in the history of flowering plants. They take their name from the magnolia family, which is one of the largest in the group. Plants in this family nearly all have woody stems, and some kinds grow into large trees. However, the group also includes herbaceous or non-woody species, both as free-standing plants and as climbers.

Some herbaceous magnoliids have highly specialized flowers. In birthworts and pipevines, for example, the flower is shaped like a flaring tube, lined with backward-pointing hairs. Flowers like these act as temporary traps for pollinating flies, which are attracted by a powerful scent. But these are an exception. Most magnoliid flowers are structurally simple, with numerous spirally arranged parts, attached separately to a central stem. Instead of sepals and petals (see p.122), they have a single layer of flaps that are similar

in colour, size, and shape, known as tepals. The fossil record shows that similar flowers existed over 100 million years ago.

COMMON CHARACTERISTICS

Magnoliids are grouped together mainly by genetic evidence, but they also share features that can be seen with a microscope, or with the naked eye. At a microscopic level, their pollen grains have a single pore. This small but significant feature links them with monocots, but distinguishes them from eudicots – by far the largest group of flowering plants – which have three-pored pollen grains.

Most magnoliids have leaves with smooth margins, and a network of branching veins. Their fruits may be soft and fleshy, or hard and cone-like, and can contain one or several seeds. Species with fleshy fruits use animals to disperse their seeds: many of them are swallowed whole and then scattered by birds. In prehistoric times, wild avocados may have been dispersed by giant ground sloths. Now that ground sloths are extinct, avocados depend on human cultivation to disperse their seeds and ensure survival.

DIVISION	ANGIOSPERMAE
CLADE	MAGNOLIIDAE
ORDERS	4
FAMILIES	18
SPECIES	10,000

DEBATE PIONEERING PLANTS

Scientists have differing opinions about the growth habit, or shape and lifestyles, of the first flowering plants. According to the “woody hypothesis”, these plants were trees or shrubs, like many magnoliids are today. However, a rival view – known as the “paleoherb hypothesis” – suggests that the first angiosperms were herbaceous, or non-woody, plants with relatively fast life cycles. This meant that they were good at colonizing disturbed ground such as riverbanks. Currently, neither view has proved decisive, although molecular analysis does favour the idea that they were woody plants.

CANELLALES

There are two families in the order Canellales: the Canellaceae and the Winteraceae. These are aromatic trees and shrubs with leathery entire leaves. The flowers in most species have both male and female parts and the fruit is a berry. The leaves and bark of some species can be used medicinally. The Winteraceae is a primitive family – its members have woody stems that contain no water-conducting vessels.

WINTER'S BARK

Drimys winteri
F: Winteraceae
Native to the coastal rainforest in Chile and Argentina, this tree has aromatic bark and leaves and fragrant flowers.

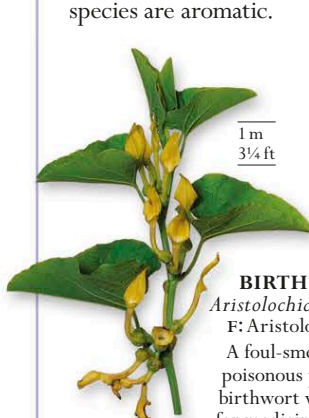


PIPERALES

The order Piperales includes three families of herbaceous plants, trees, and shrubs, widely distributed in tropical regions. The stems have scattered bundles of vascular tissue, a characteristic of monocots. Members of the family Piperaceae have tiny flowers which lack petals and are clustered in spikes. Many species are aromatic.

BIRTHWORT

Aristolochia clematitis
F: Aristolochiaceae
A foul-smelling and poisonous perennial, birthwort was grown for medicinal use. It is native to Europe, growing in damp places.



ASARABACCA

Asarum europaeum
F: Aristolochiaceae
This creeping species grows in European woodlands. Its glossy evergreen leaves hide insignificant flowers.



BLACK PEPPER

Piper nigrum
F: Piperaceae
An evergreen climber from shady habitats in S. India and Sri Lanka, this species is widely cultivated for its aromatic fruits.



MAGNOLIALES

Consisting almost exclusively of trees and shrubs, the Magnoliales are a primitive order, widely distributed in the fossil record. Although very variable, most have simple alternately arranged leaves and flowers with both male and female parts. There are six families in the order, of which the Magnoliaceae are the best known, being widely grown in gardens for their spectacular flowers.



TULIP TREE

Liriodendron tulipifera
F: Magnoliaceae

A native of eastern N. America, the tulip tree grows in woodland. The leaves turn yellow in autumn before falling.

tepals
protect
stamen

30 m
100 ft



CAMPBELL'S MAGNOLIA

Magnolia campbellii
F: Magnoliaceae

Several of the 210 or so species of Magnolia are grown as garden ornamentals. This species is native to mountain forests in China, India, and Nepal.

